



Society of Petroleum Engineers

SPE-227798-MS

Unlocking the Effect of Pore Types on Dielectric Permittivity in Carbonates

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This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

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Abstract

Dielectric permittivity measurements provide vital interpretation answers for evaluating subsurface reservoirs, including water-filled porosity, salinity, and rock texture parameters. Dielectric permittivity is sensitive to pore-scale interfacial polarization, and it provides information on the fluids in rock pores and their connectivity. Carbonate rocks are complex in nature and display a variety of pore types. This study aims to assess the effect of carbonate pore types on dielectric permittivity response and provide a reliable correlation method.

A total of 17 carbonate core samples were selected for this study. The core samples were cleaned and dried, after which their petrophysical properties, including porosity, and permeability were measured. Thin sections were prepared for all the cores, for petrographic description. Digital image analysis (DIA) was used to analyze the thin section images and to capture the different pore types and their attributes. The samples were then saturated with 30 kppm NaCl solution, and dielectric permittivity measurements were conducted on the samples at multi frequencies.

The samples exhibit wide ranges of porosity (8.6-45%) and permeability (0.3-1500 mD). The thin section photographs revealed three dominant pore types in the examined samples: moldic, interparticle, and intercrystalline pores. Normalized dielectric permittivity measurements at 10 MHz show that rocks with dominant moldic pores exhibit the highest dielectric permittivity values. Interparticle pores display the lowest normalized dielectric permittivity, while intercrystalline pores fall in between. To understand the effect of pore geometry on permittivity, DIA parameters such as perimeter over area (PoA), dominant pore size (DOMSize), and aspect-ratio (AR) were assessed. Moldic pores generally have the lowest PoA and AR, and the highest DOMSize. Intercrystalline pores show the opposite, while interparticle pores fall in between. Low PoA and high DOMSize indicate a simple pore system, while high PoA and low DOMSize indicate a complex pore network. The platy shape and simple pore system of moldic pores could explain their high dielectric permittivity. Conversely, the low dielectric permittivity of interparticle pores could be attributed to their complex pore network. These observations allow for developing a new workflow to predict carbonate pore types from permittivity dielectric measurements.

This study provides insights into the relationship between the carbonates pore types and dielectric permittivity response. For the first time, a relationship database is built between rock dielectric permittivity and DIA parameters. The results and observations allowed for the development of a workflow towards

predicting carbonate pore types from dielectric permittivity measurements, a crucial step for advanced and accurate formation evaluation.

Introduction

Carbonate rocks are complex in nature. Their texture may vary within the same rock due to post depositional diagenetic processes that alter their original texture. Pore types in carbonates are of great importance for reservoir characterization as they control fluids distribution, pathway, saturations and reservoir quality (Rahimpour-Bonab and Aliakbardoost, 2014). In carbonates, the most common porosity types are interparticle, intraparticle, intercrystalline, and moldic (Choquette and Pray, 1970). However, their effect on dielectric permittivity dispersion curves is not widely investigated. It was emphasized earlier that rock textural effects dielectric dispersion in addition to the salinity of volumetric brine in pores, and more specific at lower frequencies (Seleznev et al. 2006). They developed an analytical textural model that considers the effect of grain and pore geometries (i.e. aspect ratio) on permittivity and conductivity dispersions at frequencies between 10 MHz and 1 GHz. They found that complex refractive index model (CRIM) is not a representative model for complex geometry associated with carbonate rocks, while the textural model fits very well the experimental data and reflects variations in rock texture.

The dielectric permittivity was also used to link basic rock properties such as porosity and cation exchange capacity to develop a rock-type classification (Garrouch et al. 2009). The effect of pore microstructure and water saturation of pore systems on the dielectric permittivity was studied and it was found that the influence of secondary pores (i.e. pores created by diagenetic processes) on the dielectric permittivity depends on pore shape and decreases with oil saturation (Markov et al. 2011). In another study (Verwer et al. 2011), the effect of carbonates pore geometry parameters on electrical resistivity variations and acoustic velocities was investigated by using digital image analysis technique (DIA). The study found that the two main parameters influencing petrophysical properties are dominant pore size (DOMSize), and perimeter over area (PoA) which are obtained from thin section photomicrographs. Also, it was concluded that rocks with separated pores of moldic or vuggy porosity have the highest resistivity. Additionally, they showed that for similar ranges of porosity, samples with large pores in a simple pore structure have higher resistivity compared to samples with small pores of intricate pore structure. In a different study, a new pore classification model for carbonate rocks was developed that includes pore geometry, connectivity, and diagenesis in the pore system to characterize reservoir flow units (Tonietto et al. 2014). Recently, the equivalent pore aspect ratios (EPAR) are used as a tool to predict pore types by combining acoustic velocities, porosity, bulk density and mineralogical composition of carbonate rocks (Fournier et al. 2018). They found that correct pore type detection from acoustic and porosity measurements is largely dependent on the pore type classification. Another study used three pore shapes to simulate effects of wettability and pore geometry on electrical resistivity and capillary pressure (Man and Jing, 2000). They found that pores of star-like grain boundary shape are suitable to simulate pore-scale fluid distribution and electrical conduction.

It was noted that dielectric dispersion of bulk water alone is almost constant over a wide range of frequency (Seleznev et al. 2006) (i.e. up to 1 GHz). Several dielectric models were developed to explain permittivity dispersion behavior of carbonate rocks. At low frequency, spherical grain shapes can't explain the dielectric behavior (Feng and Sen 1985). Therefore, a model was developed to consider not only spherical geometries but also platy grain shapes that may present in partially saturated rocks and this model was called bimodal model.

Brine-saturated rock samples showed an increase in the dielectric permittivity at low frequency ranges, and this increment varied from rock to rock (Poley et al. 1978). A new proposed model with three mechanisms to explain this behavior: texture effect, effect of the double layer especially for clay-bearing rocks, and binding of water to solid surfaces (Sen 1981). Another study is conducted to investigate the effect of calcite texture on dielectric properties (Kenyon 1984). They showed that the increase in grain

platininess will result in a decrease in conductivity at low ranges of frequency. Also, they found that at high frequency (i.e. 1.1 GHz), rock samples saturated with high salinity water will show an increase in dielectric permittivity. The current research study objective is evaluating the effect of dominant pore type and pore structure attributes on dielectric response of carbonate samples fully saturated with brine. This integrated study of petrography and dielectric can contribute parameters for understanding the behavior of dielectric dispersion from the dielectric scanner tool and predicting carbonate pore types downhole. In the current study, we evaluate the effect of different carbonate pore types on dielectric permittivity and conductivity dispersion curves.

Materials and Experimental Measurements

Seventeen carbonate core samples from different outcrops are used for the study. The core samples are initially cleaned and dried to remove possible impurities from their pores. The dominant pore types in the samples are detected and defined under optical microscope then analyzed digitally utilizing an open-source software. Thin sections are cut from the end of the core samples and impregnated with blue epoxy to reveal porosity. The overall tasks of the study are shown graphically in Figure 1:

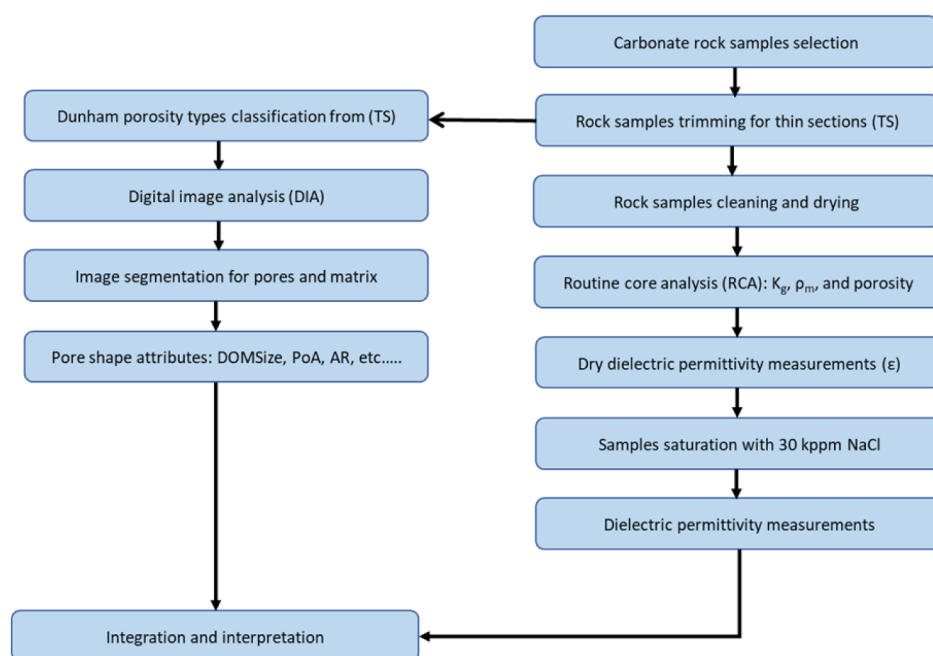


Figure 1—Flowchart for the study.

The porosity and gas permeability of the core samples are measured with AP-608 Automated Permeameter-Porosimeter at 500 psi confining pressure using helium gas in the steady-state technique.

Seventeen thin sections are prepared, and their photomicrographs are captured using Leica DM RX Microscope with MC 190 HD camera under plane polarized light (PPL). Dunham classification was used to classify texture of the selected carbonate samples. Seventeen thin sections were prepared, and their photomicrographs were captured and described to characterize the texture and pore types.

After thin section preparation, digital image analysis technique is applied for measuring quantitative pore structure parameters. The three most important parameters are: (1) perimeter over area (PoA) which is the ratio between total perimeter that encloses the pore space and the total pore space area on a thin section. (2) dominant pore size (DOMSize) which represents the upper boundary of pore sizes with which 50% of the porosity on a thin section is composed. It is an indication of the pore-size range that dominates the rock

sample. (3) aspect ratio which is the ratio between the maximum and the minimum axis of an ellipse that bounds the pore.

The dielectric setup comprises an Agilent network analyzer (ENA series E50771C), which is connected to a laptop and controlled via a program. The network analyzer is calibrated using a set of short, open, and 50 Ohm standards from Agilent. The samples are placed in a Teflon core holder and gently pressed against a coaxial probe to establish a good contact between the probe and the flat end of core samples.

It is common to express the sample dielectric properties, permittivity and conductivity, as complex dielectric permittivity:

$$\varepsilon^* = \varepsilon_r + i \frac{\sigma}{\omega \varepsilon_0} \quad (1)$$

where ε_r is the relative dielectric permittivity or the dielectric constant and is the real part of the complex permittivity. The imaginary part relates to the high frequency conductivity of the sample (σ), the free-space permittivity (ε_0), and the angular frequency (ω). Note that this is the relative complex permittivity as it is normalized relative to that of free space.

Results and Discussions

Dry Core Measurements

After cleaning and drying of the rock samples, gas porosity and permeability and dielectric permittivity are obtained. The petrophysical properties of the selected core plugs are summarized in Table 1. The dielectric permittivity data for the dry core samples are shown in Figure 2. Each core was measured from top and bottom sides. However, only the upper-side measurements are considered in this study for consistency in the upcoming comparisons. Dielectric permittivity values of the dry state range between 6.28 and 8.03.

Table 1—Petrophysical properties of the studied core samples.

No.	Dominant Pore Type	Sample ID	Lithology	Diameter [cm]	Length [cm]	Helium porosity [%]	Gas permeability [mD]
1	Interparticle	AC-1	Limestone	3.81	4.97	28.5	29
2	Interparticle	Q21-03	Limestone	3.82	4.98	25.8	83
3	Interparticle	Q23-02	Limestone	3.81	4.97	37.3	87
4	Interparticle	IND-115-1	Limestone	3.81	4.97	16.8	88
5	Moldic - elongate	Q6-07	Limestone	3.82	4.96	18.7	134
6	Moldic – Elongate	Q21-01	Limestone	3.82	4.98	31.9	54
7	Moldic – Elongate	Q21-02	Limestone	3.82	4.89	22.0	124
8	Moldic – Elongate	Q21-09	Limestone	3.81	4.99	39.5	143
9	Moldic – Elongate	Q21-10	Limestone	3.82	4.98	35.3	1468
10	Moldic – Ellipsoid	Q21-14	Limestone	3.82	4.96	45.0	72
11	Moldic – Ellipsoid	Q22-03	Limestone	3.81	5.00	34.6	17
12	Moldic- Crystal	Q10-04	Limestone	3.82	4.97	13.9	9.1
13	Intercrystalline	WN-1	Dolostone	3.82	4.95	10.1	3.90
14	Intercrystalline	DSII-3	Dolostone	3.82	4.93	9.8	129
15	Intercrystalline	A101-05	Dolostone	3.82	4.90	18.5	12.8
16	Intercrystalline	Q15-03	Dolostone	3.82	4.98	14.5	8.0
17	Intercrystalline	Q9-08	Dolostone	3.83	4.99	9.2	20

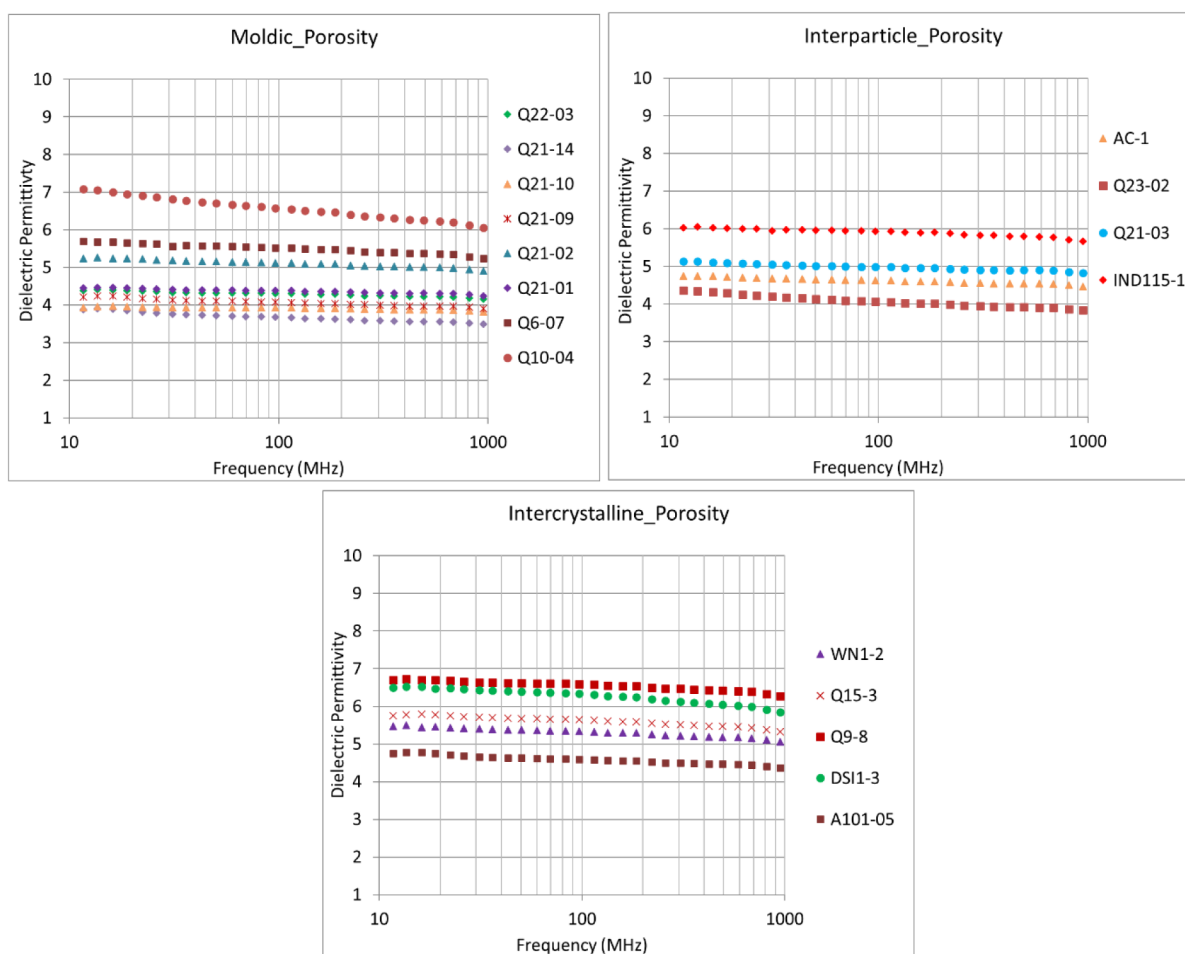


Figure 2—Dielectric permittivity for dry state of the selected samples after grouping them based on the dominant pore type.

Saturated Core Measurements

After completing routine core analysis and dielectric permittivity measurements in the dry state, all seventeen core plugs were saturated with 30 kppm NaCl brine. Saturated-state dielectric permittivity measurements were then conducted. The permittivity of the samples was initially grouped based on the dominant pore types, and results show a first-order control of frequency on dielectric permittivity behaviour with a clear dispersion as frequency increases (Figure 3). The dielectric constant of the measured samples has a dispersion starting from 10 MHz to approximately 400 MHz which is due to interfacial polarization related to rock texture (i.e. pore geometry and structure). Beyond this range, to 1 GHz, the dispersion flattens, and only molecular polarization remains active, becoming independent of texture (Abdallah et al. 2021).

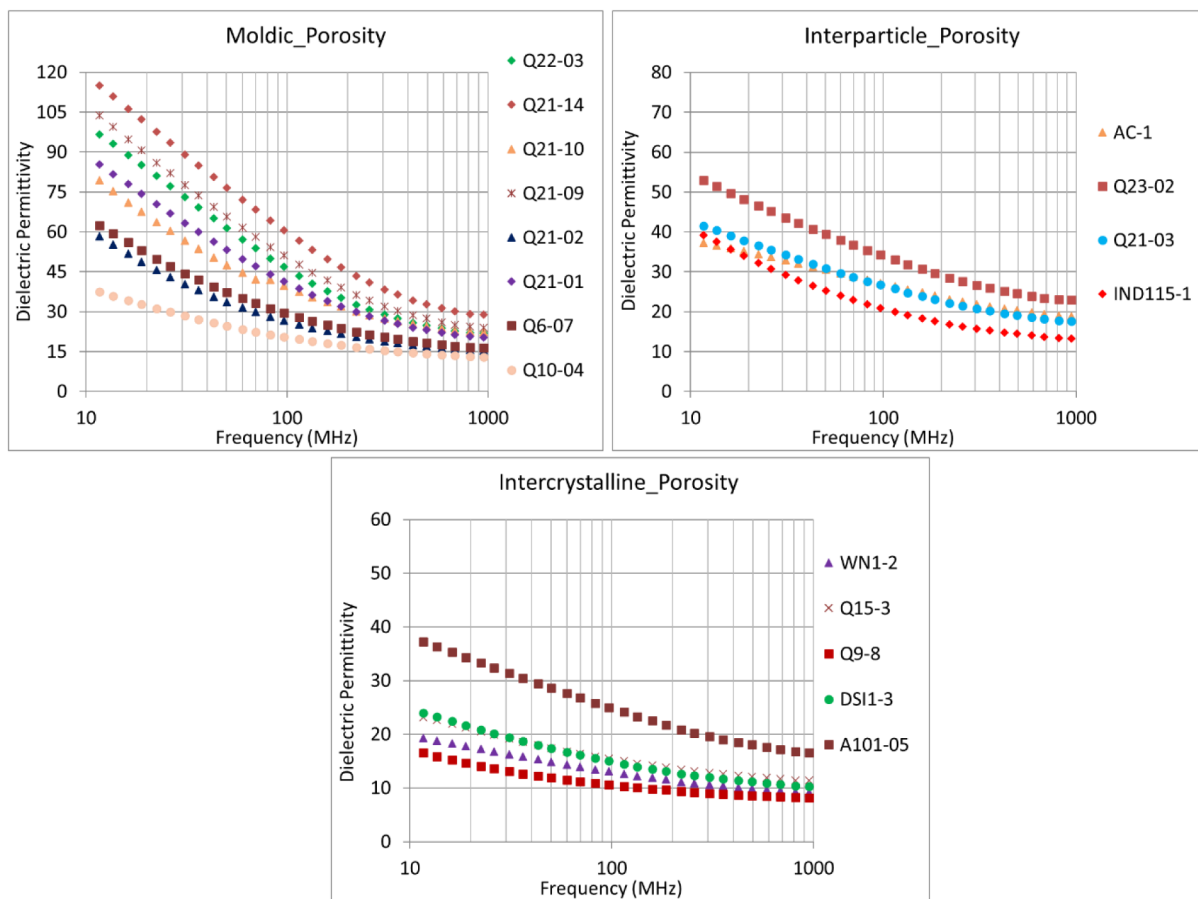


Figure 3—Dielectric permittivity of the samples saturated with 30 kppm NaCl. Dielectric permittivity results are grouped based on the dominant pore type.

Influence of Dominant Pore Types

The results show a noticeable variation in permittivity at 10 MHz (Figure 3) and (Figure 4), reflecting the influence of rock texture. This analysis emphasizes the role of pore type in driving differences in dielectric response. Based on the Complex Refractive Index Model (CRIM):

$$\varepsilon_{eff} = \left((1-\phi)\sqrt{\varepsilon_m} + \phi S_w \sqrt{\varepsilon_w} + (1-S_w)\phi \sqrt{\varepsilon_{HC}} \right)^2 \quad (2)$$

Where:

ε_m : Matrix permittivity

ε_w : Water permittivity

ε_{HC} : Hydrocarbon permittivity (If present)

ϕ : Porosity

The dielectric permittivity at 10 MHz shows noticeable variation with porosity ranges, as illustrated in Figure 4. This variation in permittivity at 10 MHz is closely linked to differences in rock texture (Seleznev et al. 2006). Rock samples with dominant moldic pores exhibit the highest permittivity values, while those with intercrystalline pores show the lowest. Interparticle pores yield intermediate values, with some overlap between them. These trends suggest that, for similar porosity ranges, pore texture and geometry play a critical role in shaping the dielectric response. Understanding these structural controls is essential for interpreting permittivity variations in carbonates. It is obvious that effective rock permittivity increases with the amount of pores present in the rock (i.e. porosity) as shown in (Figure 4A). Therefore, to isolate the effect of pore structure from that of porosity itself, permittivity values at 10 MHz were normalized by porosity. This was done by dividing the permittivity value of each sample by its corresponding porosity ($\varepsilon /$

ϕ), resulting in a porosity-independent permittivity index. This normalization allows to compare samples with different porosities on a consistent basis and focus only on how pore type (moldic, interparticle, and intercrystalline) influences the dielectric response. By controlling porosity this way, we minimize its impact and highlight the contribution of pore structure/geometry on the observed permittivity variations.

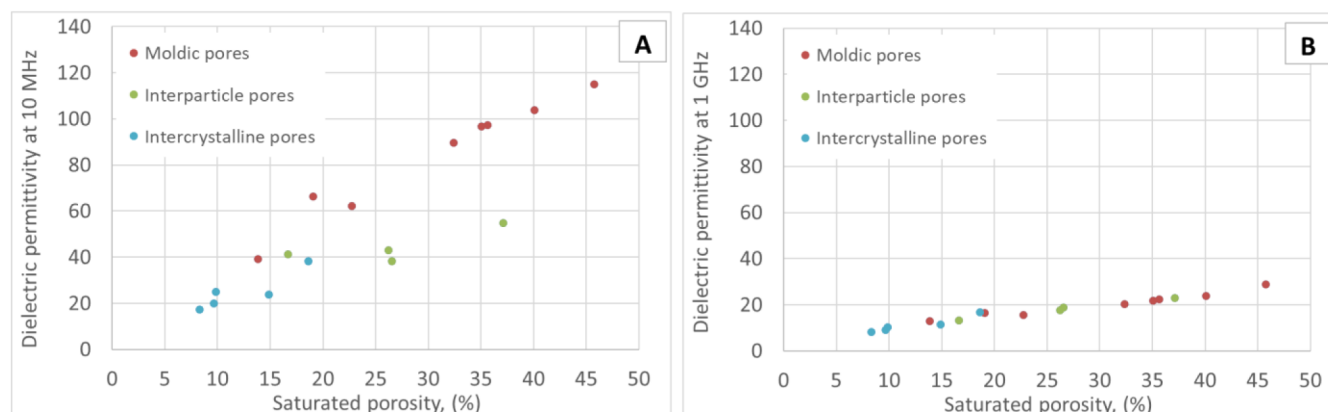


Figure 4—Dielectric permittivity with different dominant pore types at 10 MHz (A) and 1 GHz (B).
The distinction between pore types is evident at 10 MHz due to active interfacial polarization, while at 1 GHz, the dielectric response flattens and becomes less sensitive to pore structure.

As frequency increases to 1 GHz, the permittivity dispersion becomes increasingly less sensitive to pore structure. This is because polarization mechanisms such as Maxwell–Wagner and interfacial polarization, which dominate at lower frequencies, becomes less effective (Bona et al. 1998). As a result, the permittivity curve flattens at high frequencies, and differences between samples with varying pore structures become less distinct (Figure 3), and (Figure 4b).

Dielectric permittivity and digital image analysis (DIA)

Dielectric measurements of the saturated rock samples show a first-order control of frequency on dielectric permittivity behaviour with a clear dispersion as frequency increases (Figure 5). The dielectric constant of the measured samples has a dispersion starting from 10 MHz to approximately 400 MHz which is due to interfacial polarization related to rock texture (i.e. pore geometry and structure). Beyond this range, at 1 GHz, the dispersion flattens, and only molecular polarization remains active, becoming independent of texture (Abdallah et al. 2021).

At 10 MHz, normalized permittivity values range between 1.5 to 3.5, with pore texture playing a significant role in this variation. Thin-section photomicrographs reveal a diversity of pore types, moldic (vuggy), interparticle, and intercrystalline. Dissolution features, along with cement distribution, influence both porosity and permittivity. Moldic pores consistently show the highest normalized permittivity, while interparticle pores exhibit the lowest. Further analysis identifies three dominant pore geometries among the sample: moldic ellipsoid, platy (Figure 5A), moldic round (Figure 5B), moldic elongate (Figure 5C), as well as intercrystalline (intergranular) (Figure 5D), and intercrystalline (Figure 5E). Moldic pores, in particular, are often associated with platy grain textures and equant calcite crystals, contributing to their elevated dielectric values. This observation is consistent with earlier findings (Sen, 1981; Kenyon, 1984) linking platy grains to higher permittivity. Among the pore types, interparticle pores tend to exhibit lower normalized permittivity than intercrystalline pores. To further understand how pore geometry controls dielectric response, digital image analysis (DIA) is performed. Parameters such as PoA, DOMSize, and aspect ratio are quantified and correlated with permittivity variations across the different pore types.

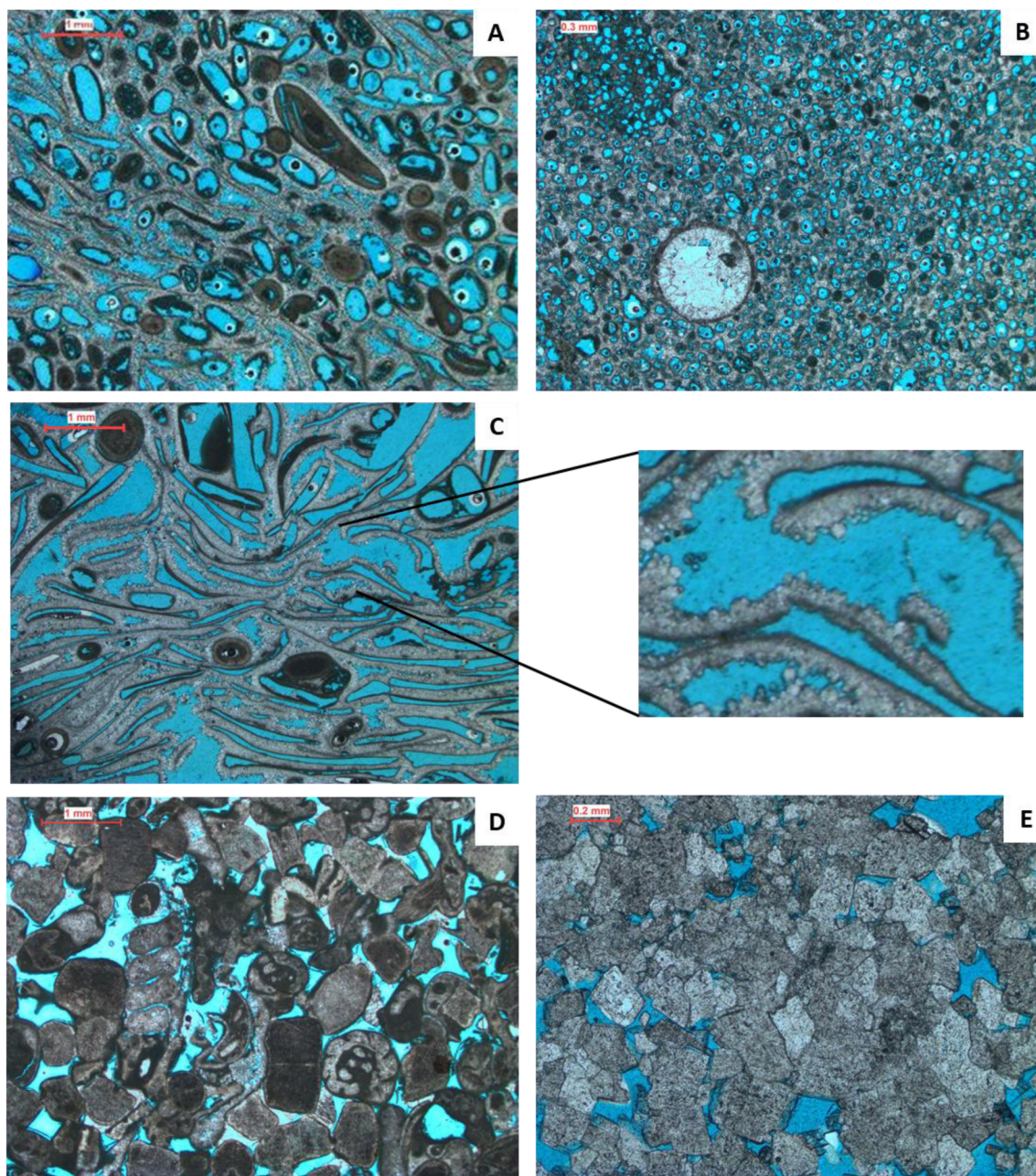


Figure 5—Main pore types in the studied samples. (A) Moldic-ellipsoid (platy grains and pores). (B) Moldic-round. (C) Moldic-elongate. (D) Interparticle. (E) Intercrystalline.

Figure 6 A-C shows crossplots of the dielectric permittivity results versus the DIA parameters with the identified pore types are superimposed in color. In the analysed thin sections, the values of perimeter over area (PoA) range from 30 to 290, while the dominant pore size (DOMSize) values fall between 21 and 493, and the aspect ratio values vary from 0.31 to 0.59. Overall, the moldic pores display the lowest PoA and aspect ratio, and the highest DOMSize. On the contrary, the intercrystalline pores show the highest PoA and aspect ratio, and the lowest DOMSize. The DIA parameters of the interparticle pores fall between the moldic and intercrystalline values. High PoA and low DOMSize indicate intricate pore network, while low PoA, low aspect ratio and high DOMSize indicate simple pore network with larger pores (Verwer et al. 2011). It is obvious from the above that intricate pore network would result in lower dielectric permittivity, while simple pore network and larger pores would result in high dielectric permittivity. Large moldic pores are found to be associated with tiny crystal cements covering internal and external parts of the pores (Figure 5

C), that may increase the available surface area which allows more accumulations of free electrical charges exist in the pore fluid and hence increases permittivity and vice versa.

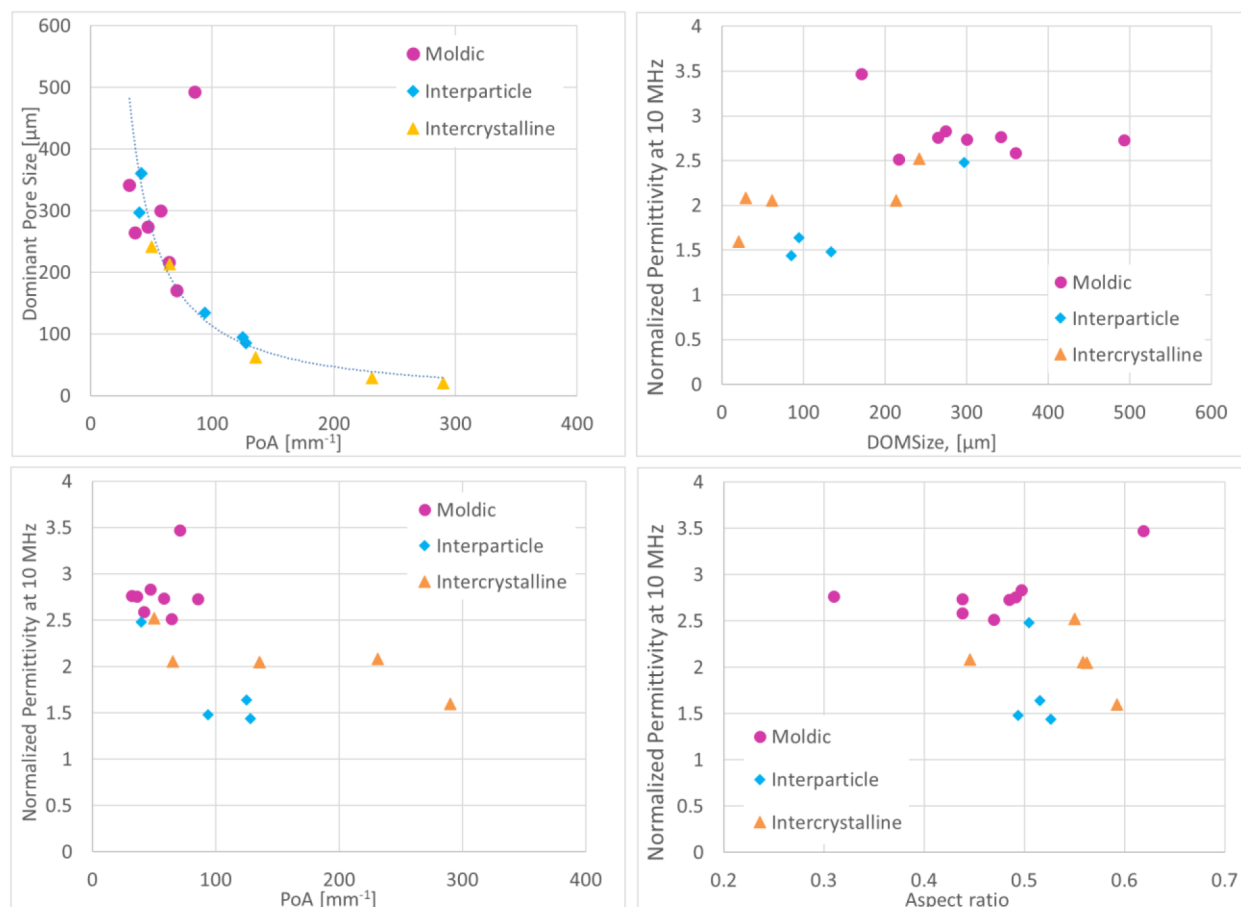


Figure 6—Crossplots show the relation between dielectric permittivity and digital image analysis parameters with the dominant pore types superimposed in color.

Based on the permittivity ranges of the measured samples, it is possible to predict carbonates pore types from dielectric dispersion data measured downhole by the Dielectric Scanner Tool (ADT). However, further investigation on the dielectric permittivity of the interparticle and intercrystalline dominant pores as well as the effect of DIA parameters on physical behavior of dielectric permittivity still needed, as some overlaps were observed.

Conclusion

The examined carbonate rocks in the current study show a remarkable change in normalized dielectric permittivity. Due to the fact that dielectric permittivity is highly affected by rock texture at relatively low frequency ranges (10 MHz), detailed pore typing, and pore-structure quantification of the studied samples is conducted utilizing thin section petrography. Three dominant pore types are observed: moldic, interparticle, and intercrystalline. Considering the normalized permittivity, it is found that moldic pores display the highest dielectric permittivity, while interparticle pores show the lowest. Pore-structure parameters are quantified through running digital image analysis (DIA) on thin sections. The quantitative DIA parameters, PoA, DOMSize, and aspect ratio show a good correlation to variations in the normalized dielectric permittivity and dominant pore types. Samples with dominant moldic pores (high dielectric permittivity) have simple pore network with larger pores, that is, low PoA and aspect ratio, and high DOMSize. The large size of moldic pores and their platy shape (except for few samples) may allow more electrical charges

to accumulate on their pore surfaces, thus increasing their dielectric permittivity. On the contrary, samples with dominant interparticle pores (low normalized dielectric permittivity) have intricate pore system and thus intermediate PoA, aspect ratio, and DOMSize. Intercrystalline pores show, however, show intermediate normalized permittivity, but the highest overall the lowest DOMSize, and the highest PoA and aspect ratio. The current work sheds light on the importance of pore texture parameters to explain the dielectric permittivity behavior in carbonate rocks and thus make it possible to predict the carbonate pore types from dielectric logging tools. Our hypothesis from this research study is as shown in Figure 8, where carbonates pore types can be predicted from dielectric dispersion data measured downhole by ADT. However, further investigation on the dielectric permittivity of the interparticle and intercrystalline dominant pores as well as the effect of DIA parameters on the dielectric behaviour is still needed, as some overlaps were observed.

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